Ruipu Shi

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ABOUT ME

Data Engineer with hands-on experience building data ingestion, integration and quality pipelines across web, API, government and geospatial data sources. Skilled in Python, SQL and relational databases, with practical experience transforming raw data into reliable, database-ready datasets. Experienced in data normalisation, entity resolution, geospatial integration, automated monitoring and scalable data-processing workflows.

KEY STRENGTHS

Team Collaboration & Leadership

Analytical & Problem-Solving Mindset

Communication & Stakeholder Engagement

Adaptability & Initiative

Attention to Detail & Quality Assurance

WORK EXPERIENCE

Urban Intelligence, Canberra
Aug 2026 - Present

Junior Data Analyst Intern

- Investigated and integrated undocumented ACT development application datasets from multiple nested JSON and GeoJSON sources, reconstructing relationships between datasets and building repeatable workflows for data normalisation and internal database ingestion.
- Developed geospatial data integration workflows using coordinates and point/polygon geometries to associate government land and development records with internally defined parcel polygons and identifiers for consistent cross-source indexing.
- Built a document lifecycle monitoring framework for nearly 300 external documents, implementing structured source/version tracking, automated change detection and rule-based successor identification, with manual review for ambiguous cases.

QX Tech, Canberra
Nov 2025 - Present

Data Engineer Intern

- Built automated data collection and enrichment pipelines using Python and SQL to process large-scale company datasets from multiple web and API sources, transforming unstructured information into structured records for downstream applications.
- Developed data quality and entity-resolution workflows to identify missing, duplicated and inconsistent records, combining deterministic validation rules with automated extraction and enrichment to improve data completeness and reliability.
- Designed structured ingestion workflows for database-ready company records, supporting evolving schemas and downstream analytics and application requirements.
- Applied machine learning to business classification tasks, using confidence-based thresholds to automate reliable predictions while separating uncertain cases for manual review.

EDUCATION

Master of Data Science
University of Canberra
July 2025 - Present

Bachelor of Computing
Australian National University
July 2020 - Dec 2024

TECHNICAL SKILLS

Programming	Python (Pandas, NumPy, Scikit-learn), SQL, R
Data Engineering	ETL Pipelines, API & Web Data Ingestion, Data Integration, Data Transformation, Data Validation, Data Quality Management, Entity Resolution
Databases	PostgreSQL, Relational Databases, Schema Design, Data Migration
Analytics & Visualisation	Power BI, Tableau, Excel
Tools & Platforms	Git, Linux, AWS (EC2, S3, VPC)

ACADEMIC & TECHNICAL PROJECTS

Company Enrichment Pipeline | Data Engineering Project
Feb 2026 -Apr 2026

- Designed and implemented a data processing pipeline to transform unstructured web data into standardised company records.
- Built automated cleaning, validation, and deduplication workflows using Python and SQL.
- Applied rule-based data quality checks to resolve conflicting information across multiple sources.
- Improved dataset completeness and consistency, producing reliable outputs for downstream applications and analytics.
- Processed and enriched over 30,000 company records through automated data transformation and validation workflows.

Pet Image Classification System | Computer Vision
Mar 2026

- Built an image classification pipeline using PyTorch, fine-tuning a pre-trained ResNet18 model for binary classification
- Applied data preprocessing and augmentation techniques (e.g., normalization, transformations) to improve model generalisation
- Evaluated model performance using validation metrics and confusion matrix, achieving ~94% accuracy on unseen data

Executive Analytics Dashboard | ANU Data Transformation Initiative
Feb 2024 – Nov 2024

- Designed and prototyped a high-fidelity executive dashboard to address ANU's internal data management challenges.
- Conducted stakeholder interviews, synthesised user insights, and built interactive visualisations using Power BI, Python, and Miro.
- Earned a Distinction, showcasing strengths in data storytelling and user-centred design.

Reference available upon request.